

WIND PRESSURE

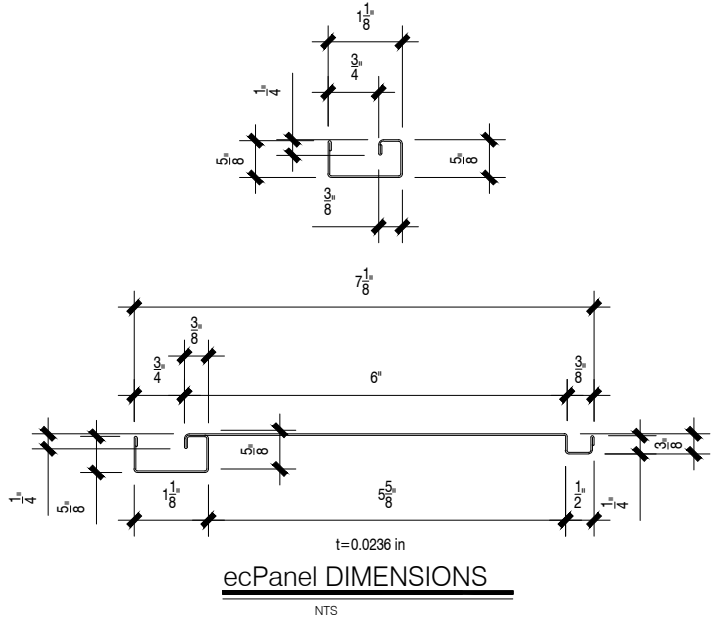
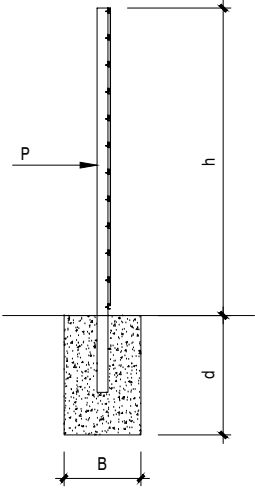
PER ASCE 7-22, SECTION 26.10-1
WIND VELOCITY V= 115 MPH EXPOSURE C
Kzt= 1
Kh= 0.85
Kd= 0.85
G= 0.85

BASIC WIND PRESSURE: qz= 0.00256 Kzt Kh Kd V^2 0.6
qz= 14.6766 PSF

PER ASCE 7-22 SECT 29.3 FIG 29.3-1

FENCE DIMENSIONS h= 6 B= 16
s= 6 s/h= 1 B/s= 2.66667
THEN Cf= 1.4

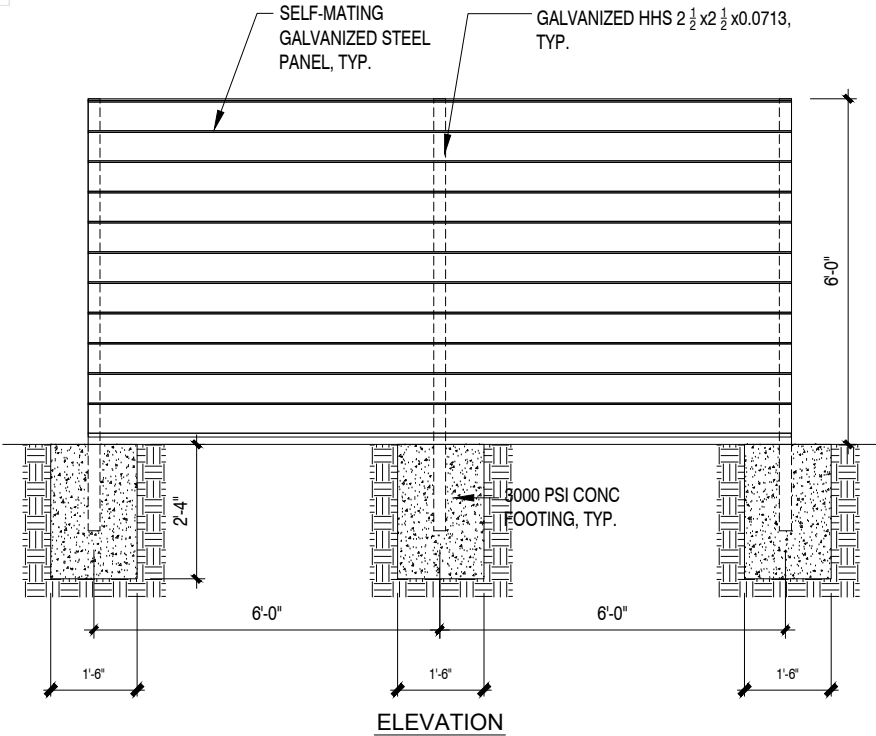
DESIGN WIND LOAD F=qz G Kd Cf F= 14.8454 PSF



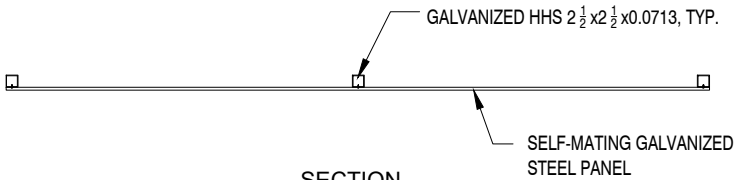
MINIMUM FENCE FRAME FOR V=115 MPH				
WIND EXPOSURE	POST SIZE HSS	SLAT THICKNESS (INCH)	FOUNDATION SIZE	HORIZ STRUT HSS
"C"	2 1/2 x 2 1/2 x 0.0713	0.0236	18"Øx2'-6" DEPTH	2x1x0.0713

STRESS CAP: Fb=30.36 ksi

FENCE HEIGHT UP TO 6'-0"



ELEVATION

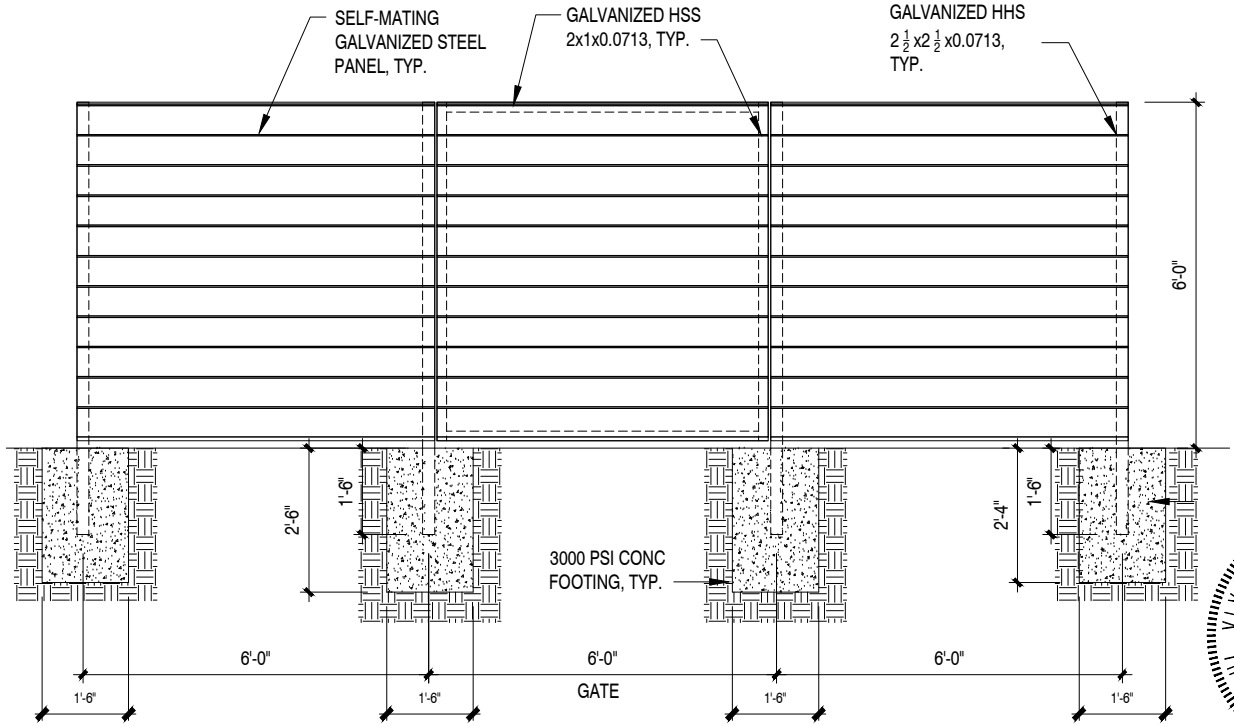


SECTION

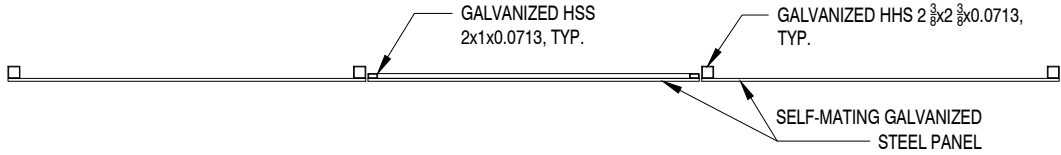
TYPICAL FENCE

NTS

1



ELEVATION

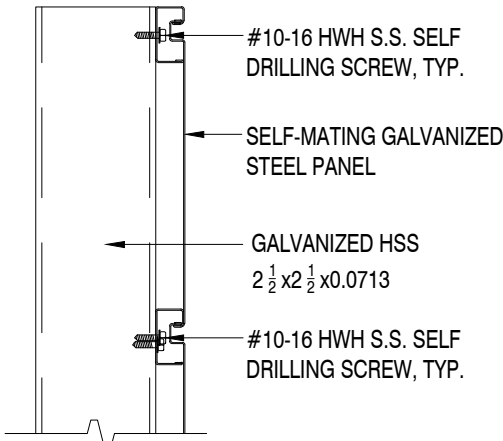


SECTION

TYPICAL FENCE W/ GATE

NTS

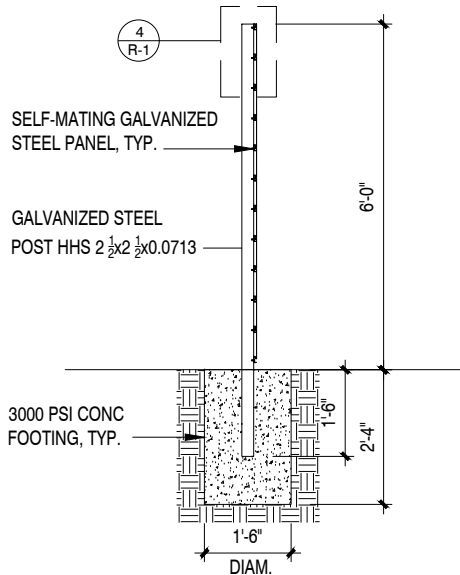
2



CONNECTION DETAIL

NTS

4



TYPICAL FENCE SECTION

NTS

3

GALVANIZED FENCE POST FOOTING

Wind pressure	F= 14.84 psf	Trib= 6	ft	w= FxTrib
Calculate Depth Embedment	FOR EXPOSURE "C"			w= 89.04 plf
Height above ground line at which force P is applied, h=	3	ft		Post height 6 ft
Total horizontal load	P= 534.24	lbs		P= 534.24
Diameter of concrete casing, B=	1.5	ft		
So=	400 psf (form FBC 2023, Table 1806.2)			
Assume d1=	2.55 ft			and movement of pole=1/2" Use So 2 times per 1806.3.4
S1=So*d1^2/3=	680.0 psf			
A=2.34*P/(S1*B)=	1.2			
d=A/2*(1+√(1+(4.36*h/A)))=	2.71	ft		

00200 BUILDING CODES

FLORIDA BUILDING CODE - 2023 8th EDITION

00500 STRUCTURAL LOADING:

THE STRUCTURE HAS BEEN DESIGNED IN ACCORD WITH THE 2023 FLORIDA BUILDING CODE AND / OR MORE RESTRICTIVE REQUIREMENTS FOR LOADS AS GIVEN BELOW UNLESS SPECIFICALLY CALL FOR DIFFERENT LOADING CRITERIA. REFER TO DRAWINGS FOR LOAD SCHEDULE.

WIND LOAD PER ASCE 7-22:
V=115 MPH FOR FENCES UP TO 6'-0"

FOUNDATIONS:

* PERFORMED: WE OBSERVED A SANDY SOIL WITH LIMESTONE FRAGMENTS. AS PER FBC 2023; TABLE 1806.2; SECTION 1806.3.4 WE ASSUME A MAXIMUM OF 200 PSF LATERAL SOIL BEARING PRESSURE CAPACITY (WILL BE INCREASED BY 2)

03300 CAST-IN-PLACE CONCRETE: ACI 318-11

ALL CONCRETE TO HAVE 28 DAY COMPRESSIVE STRENGTHS AS
F'C = 3,000 PSI

05120 STRUCTURAL STEEL: AISC 13TH EDITION

PER LATEST EDITION OF THE AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC) "SPECIFICATIONS FOR THE DESIGN, TO CONFORM TO:

SHAPES & PLATES

ASTM A 500 GRADE C
FY=36 KSI; FB=23.76 KSI

TUBES

ASTM A 500 GRADE B

PROJECT NAME: Metal Fence

MANUFACTURER: ecFENCE SYSTEMS

ADDRESS: FOR USE IN THE HVHZ FLORIDA BUILDING CODE 2023

Date 5/20/2025

Scale NOTED

Drawn MDH

Job ecFENCE

Sheet

S-1

1 OF 1

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